

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 9: Some Applications of Trigonometry

Previous year board questions, topic-wise: 16 questions

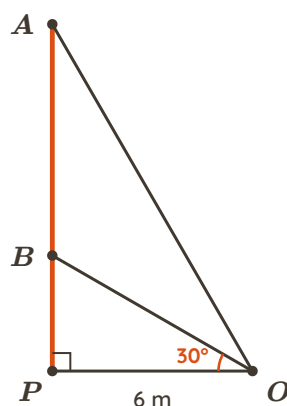
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 One right-angled triangle

1. A car drives away from the foot of a tower 30 m high. What is the angle of elevation of the top of the tower from the car when it is $10\sqrt{3}$ m from the foot? (a) 30° (b) 45° (c) 90° (d) 60°
CBSE 2026, 1 mark
2. A kite is flying 150 m above the ground. Its string is straight and makes an angle of 30° with the ground. How long is the string? CBSE 2025, 1 mark
3. A ladder leaning against a wall makes an angle of 60° with the ground. Its foot is 2.5 m from the wall. Find the length of the ladder. CBSE 2016, 1 mark

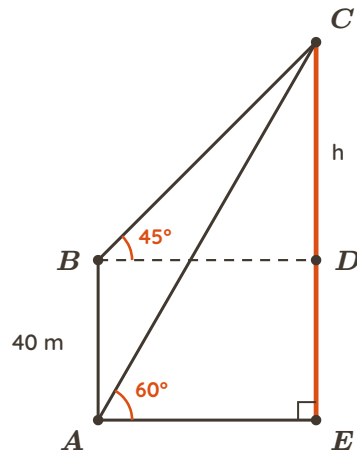
2 Two right-angled triangles

4. A radio tower PA is built in two sections: PB is the lower section and BA the upper one. Wires from a point O on the ground, 6 m from the foot P , hold the tower at B and at A . From O the angle of elevation of B is 30° and that of A is 60° . (i) Find the length of the wire OB . (ii) Find the length of the wire OA . (iii) Find AB , OR find the area of $\triangle OPB$. CBSE 2026, 4 marks



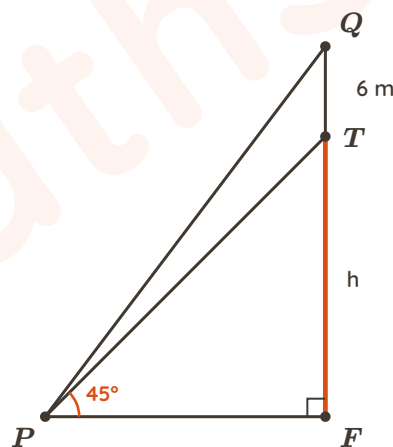
Angle $POB = 30^\circ$ and angle $POA = 60^\circ$

5. From a point A on the ground, the top C of a lighthouse CE is seen at an angle of elevation of 60° . From a deck at B , 40 m directly above A , the angle of elevation of C is 45° . D is the point on CE level with B , and $CD = h$ metres. (i) Write BD in terms of h . (ii) Write BC in terms of h . (iii) Find the height CE , taking $\sqrt{3} = 1.73$, OR find AE if $AC = 100$ m. CBSE 2025, 4 marks



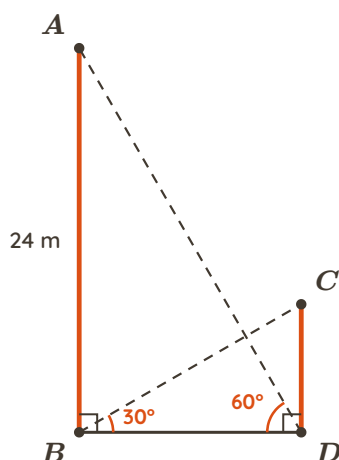
6. A 6 m pole stands on top of a tower. From a point P on the ground, the top of the pole is seen at an angle of elevation of 60° . From the top of the tower, P is seen at an angle of depression of 45° . Find the height of the tower and the distance of P from its foot. (Take $\sqrt{3} = 1.73$.)

CBSE 2024, 5 marks

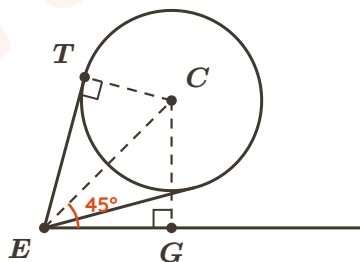


FT is the tower and TQ is the pole

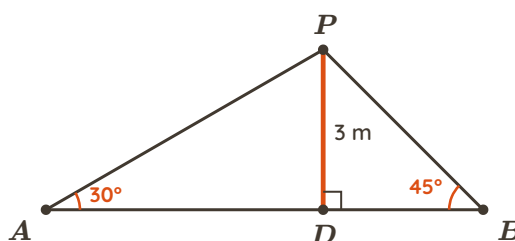
7. Two towers stand on level ground. The first is 24 m high, and from the foot of the second tower its top is seen at an angle of elevation of 60° . From the foot of the first tower, the top of the second is seen at an angle of elevation of 30° . Find the distance between the towers, the height of the second tower, and the length of a wire joining the two tops. **CBSE 2023, 5 marks**



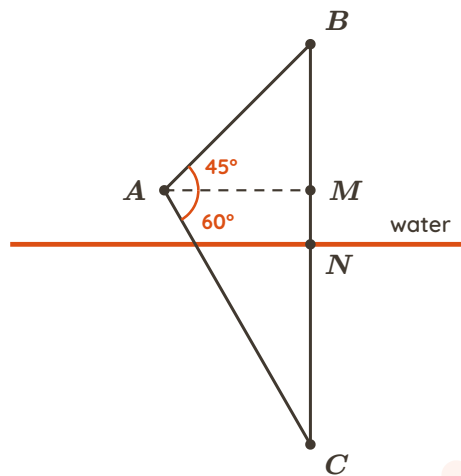
8. A spherical balloon of radius r subtends an angle of 60° at the eye of an observer, and the angle of elevation of its centre from the eye is 45° . Prove that the centre of the balloon is at a height of $\sqrt{2}r$. **CBSE 2023, 5 marks**



9. A tower is 50 m high. From the foot of a building, the top of the tower is seen at an angle of elevation of 60° . From the foot of the tower, the top of the building is seen at an angle of elevation of 30° . Find the height of the building. **CBSE 2022 Term 2, 3 marks**
10. From a point on a bridge 3 m above the banks of a river, the two banks on opposite sides are seen at angles of depression of 30° and 45° . Find the width of the river. **CBSE 2022 Term 2, 3 marks**

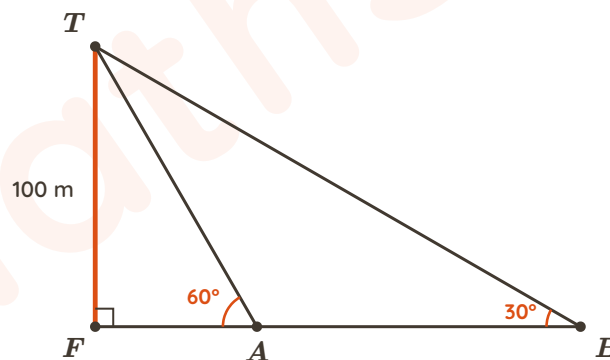


11. The top B of a pavilion standing in a lake is 10 m above the water. From a point A , h metres above the water, the angle of elevation of B is 45° and the angle of depression of the reflection C of B in the water is 60° . (i) Draw a labelled figure. (ii) Find h , taking $\sqrt{3} = 1.73$. **CBSE 2022 Term 2, 4 marks**

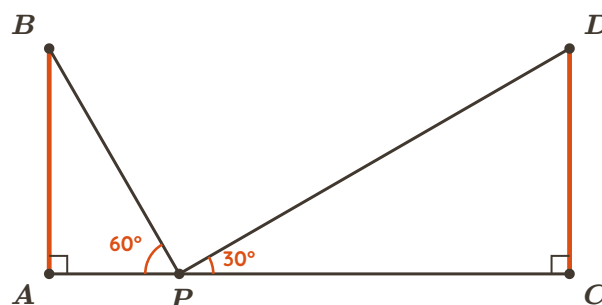


N is on the water surface, with $BN = NC = 10$ m

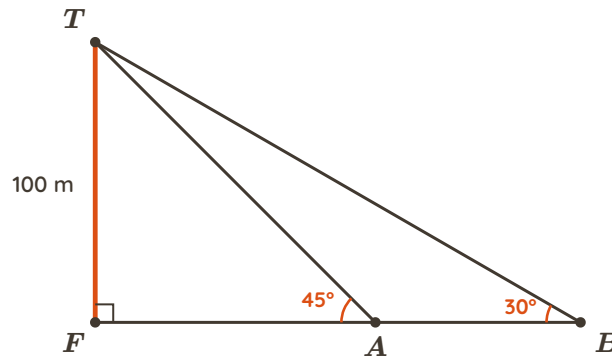
12. A man rows a boat directly away from a lighthouse 100 m high. In 2 minutes the angle of elevation of the top of the lighthouse changes from 60° to 30° . Find the speed of the boat in metres per minute. Take $\sqrt{3} = 1.732$. **CBSE 2019, 4 marks**



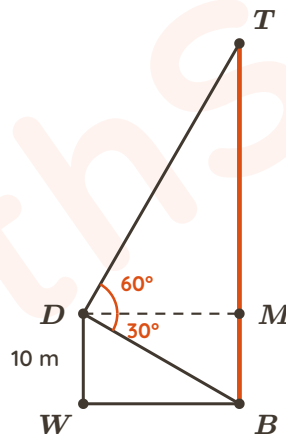
13. Two poles of equal height stand opposite each other on the two sides of a road 80 m wide. From a point on the road between them, the tops of the poles are seen at angles of elevation of 60° and 30° . Find the height of the poles and the distances of the point from them. **CBSE 2019, 4 marks**



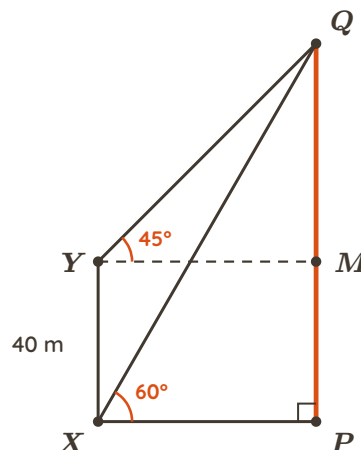
14. From the top of a lighthouse 100 m above sea level, two ships are seen at angles of depression of 30° and 45° . One ship is directly behind the other on the same side of the lighthouse. Find the distance between the ships. Take $\sqrt{3} = 1.732$. **CBSE 2018, 4 marks**



15. A man stands on the deck of a ship, 10 m above the water. He sees the top of a hill at an angle of elevation of 60° and the foot of the hill at an angle of depression of 30° . Find the distance of the hill from the ship and the height of the hill. **CBSE 2016, 3 marks**



16. The top Q of a vertical tower PQ is seen at an angle of elevation of 60° from a point X on the ground. From a point Y , 40 m vertically above X , the angle of elevation of Q is 45° . Find the height PQ and the distance PX . Take $\sqrt{3} = 1.73$. **CBSE 2016, 4 marks**



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